

Cadence — Demo Run Card

Freeze-of-gait monitor · one ankle accelerometer · on-body OLED + wireless dashboard



RUN IT — in this order

- 1 Power both boards: CPX on its LiPo, ESP32 on a USB power bank. The CPX OLED shows the CADENCE splash.
- 2 On the laptop, join Wi-Fi Cadence-Hub (password cadence123), then open <http://192.168.4.1/>
- 3 Stand still and press button A on the CPX. "CALIB" shows for ~4 s, then the movement floor is set.
- 4 Walk normally → green LED ring + "WALKING" on the OLED + a live accel trace on the dashboard.
- 5 Provoke a freeze: buzz the foot fast in place — don't step → red ring + inverted "FREEZE", and freezes: N ticks up.



ON THE BODY (CPX)

- **OLED, big:** WALKING / STILL / FREEZE
- FI x.xx / 1.82 (ratio vs threshold)
- E energy / floor (the movement gate)
- freezes: N (confirmed-onset count)
- LED ring: green = walk, teal = still
- blue = calibrating, red = FREEZE



WIRELESS (ESP32)

- **Hotspot:** Cadence-Hub
- Password: cadence123
- **Open:** <http://192.168.4.1/>
- Dashboard: trace, freeze state, HR/SpO2
- ESP32 LED: solid = browser on, blink = wait
- Finger on the MAX30102 for HR/SpO2



IF IT MISBEHAVES

OLED blank — check SDA=A5 / SCL=A4 and a shared GND; some panels are at 0x3D, not 0x3C.

Dashboard won't load — be joined to Cadence-Hub (not your usual Wi-Fi); it has no internet, which is normal.

Stuck on STILL / never FREEZE — a clean stop reads STILL — buzz the foot fast in place; if still nothing, re-calibrate (button A while truly still).

FREEZE while standing still — re-calibrate; the movement floor is set too low for this wearer/surface.

HR / SpO2 blank — rest a fingertip on the MAX30102 and hold still ~10 s for a valid reading.